DOOB'S PROOF OF MLE CONSISTENCY

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Abstract. We present a concise, self-contained exposition of Doob's proof of strong MLE consistency using surprisal and relative information. We clarify historical entropy terminology and explicitly verify all assumptions for the multivariate normal distribution using moment-generating functions and elementary matrix algebra.

1. Introduction

The consistency of Fisher's maximum likelihood estimator (MLE) [10] is a basic result discovered more than a century ago. This method is typically treated as a special case of M-estimators, and derived via standard techniques dating back to Cramér [6] and Wald [25]. A modern treatment is van der Vaart [24]. Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found Doob's consistency proof [9] of MLE convergence more motivated and accessible to students, particularly for students already familiar with information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter θ spaces and general state X spaces with no extra effort only adds to its appeal.

The contribution of this note is expository: it gives a formulation of Doob's consistency argument in surprisal/information terminology, emphasizing the role of properness and separability, and works out the multivariate normal example in detail. The proofs are designed to be self-contained throughout: in particular, the consistency argument and the matrix lemmas are developed directly, without relying on determinant multiplicativity, eigenvalue continuity, or other standard results that can obscure the elementary structure of the argument. The lemmas themselves are useful independently of their application to maximum likelihood.

We start with the setup and assumptions, then state and prove consistency. After that, we give a historical account of entropy terminology, which is an interesting story in itself, then we end by showing the assumptions hold for the multivariate normal case.

2. Surprisal and Information

A random variable X is sampled repeatedly, resulting in independent and identically distributed copies $X_1, X_2, \dots$ of X. We assume the density of X is $p(X,\theta^*)$, one of a parametric family $p(X,\theta)$ of densities, and we wish to recover θ^* from the samples $X_1, X_2, \dots$.

Stated in this manner, all we have to work with are two items: the true expectation E_* and the family $p(X,\theta)$. While $p(X,\theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over or knowledge of E_*. Given this, how do we use the samples $X_1, X_2, \dots$ to recover θ^*?

Define the *surprisal* [19, 22]

$$S(X,\theta) = \log(1/p(X,\theta)).$$

This terminology is intuitive: low probability equals high surprisal. Then it is natural to characterize θ^* as a minimizer of the expected surprisal $E_*(S(X,\theta))$. More precisely, we say θ^* is the *true parameter* if $S(X,\theta)$ is integrable for all θ and θ^* is the *unique* minimizer of $E_*(S(X,\theta))$. In particular, if θ^* is the true parameter and $\theta \neq \theta^*$, then $p(X,\theta) \neq p(X,\theta^*)$.

Equivalently, given θ^*, the *relative information* is

$$I(\theta^*,\theta) = E_*(S(X,\theta)) - E_*(S(X,\theta^*)). \tag{1}$$

The relative information is the excess expected surprisal of θ over θ^*. Then θ^* is the true parameter iff $I(\theta^*,\theta) \geq 0$ for all θ, with $I(\theta^*,\theta) = 0$ only if $\theta = \theta^*$. The relative information I appears in many fields, in different forms. Because of widely differing terminology, below we give an account of its history.

A special case is when there is a background expectation E such that[1]

$$E_\theta(f(X)) = E(p(X,\theta)f(X))$$

is an expectation for all θ. When this holds and $E_{\theta^*} = E_*$, the relative information takes the familiar form

$$I(\theta^*,\theta) = E\left(p(X,\theta^*)\log\left(\frac{p(X,\theta^*)}{p(X,\theta)}\right)\right),$$

$I(\theta^*,\theta) \geq 0$ for all θ, and $I(\theta^*,\theta) = 0$ implies[2] $p(X,\theta) = p(X,\theta^*)$ [7, 8]. When $\theta \neq \theta^*$ implies $p(X,\theta) \neq p(X,\theta^*)$, it follows θ^* is the true parameter. Usually this hypothesis is stated separately as *identifiability*. Here for simplicity it is incorporated into the definition of true parameter.

The simplest example is as follows. A vector $\theta = (\theta_1,\theta_2,\dots,\theta_d)$ is a probability vector if its components are nonnegative and sum to one. Suppose $\theta^* = (\theta^*_1,\theta^*_2,\dots,\theta^*_d)$ is a probability vector and a random variable X satisfies $P_*(X = i) = \theta^*_i$, $i = 1,2,\dots,d$. Then the mass function of X is $p(X,\theta^*) = \theta^*_X$. If θ is any other probability vector, the surprisal is $S(X,\theta) = \log(1/\theta_X)$, and

$$I(\theta^*,\theta) = \sum_{i=1}^{d} \theta^*_i \log(\theta^*_i/\theta_i). \tag{2}$$

Turning back to the general setting, the *likelihood* and (empirical) *mean surprisal*[3] are

$$L_n(\theta) = \prod_{k=1}^{n} p(X_k,\theta), \qquad \bar{S}_n(\theta) = \frac{1}{n}\log(1/L_n(\theta)) = \frac{1}{n}\sum_{k=1}^{n} S(X_k,\theta).$$

The mean surprisal is also called the negative log-likelihood.

By definition, θ^* minimizes $E_*(S(X,\theta))$. By the law of large numbers (LLN), $\bar{S}_n(\theta)$ converges to $E_*(S(X,\theta))$ as $n \to \infty$, with probability one. Hence it is natural to estimate θ^* by a minimizer of $\bar{S}_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ.

Under **A** and **B** below, with probability one, an MLE sequence $\hat{\theta}_n$ exists for sufficiently large n. In what follows, only E_* and $p(X,\theta)$ enter, E and E_θ play no role.

3. Consistency

In addition to the existence of the true parameter, we make two assumptions. A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c. Since $\hat{\theta}_n$ should minimize $\bar{S}_n(\theta)$, this leads us to the first assumption

A. *(Properness):* There is an $N \geq 1$, depending on the samples, and a proper $F(\theta)$, not depending on the samples, such that, with probability one,

$$F(\theta) \leq \bar{S}_n(\theta), \qquad \text{for all } \theta \text{ and } n \geq N. \tag{3}$$

We will need the LLN convergence $\bar{S}_n(\theta) \to E_*(S(X,\theta))$ across θ, in the sense

$$\bar{S}_n(\theta) \to E_*(S(X,\theta)), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta, \tag{4}$$

with probability one. For this we need the second assumption

[1] While $E_\theta(1) = 1$, E need not satisfy $E(1) = 1$, $E(1) = \infty$ is allowed.

[2] Apply Jensen's inequality with the strictly convex $\log(1/p)$.

[3] By analogy with mean loss in statistical learning.

B. *(Continuity):* The parameter space is separable and there is a continuous $\delta(\theta, \theta')$ and an integrable nonnegative $g(X)$ satisfying

$$|S(X,\theta) - S(X,\theta')| \leq \delta(\theta,\theta')g(X) \qquad \text{and} \qquad \delta(\theta,\theta) = 0. \tag{5}$$

Under **B**, a standard separability argument, given in the appendix, establishes (4) with probability one. Under **B**, the mean surprisal is continuous in θ. Under **A**, every sublevel set of the mean surprisal is contained in a compact sublevel set of $F(\theta)$, and hence is compact. Thus the mean surprisal is proper. Since a proper continuous function has a minimizer, an MLE $\hat{\theta}_n$ exists for $n \geq N$, with probability one.

A and **B** are particularly tractable for *exponential families* [5, 16, 20, 24], the densities $p(X, \theta)$ where $S(X, \theta)$ is a sum of products of the form $f(\theta)g(X)$.

In (4), there is no presumption of uniform convergence. This is a key feature of Doob's proof: no uniform convergence of the mean surprisal is required. The argument instead combines properness, a point-wise LLN, and the continuity estimate in (5) to control limit points of the MLE sequence.

Theorem (Consistency)**.** *Let $X_1, X_2, \ldots$ be independent and identically distributed, and let $p(X, \theta)$ be a parametric family with true parameter θ^*. Under **A** and **B**, with probability one, an MLE sequence $\hat{\theta}_n$ may be defined for sufficiently large n, and, for any MLE sequence, $\hat{\theta}_n$ converges to θ^* as $n \to \infty$.*

Proof. By the LLN,

$$\frac{1}{n}\sum_{k=1}^{n} g(X_k) \to E_*(g(X)), \qquad \text{as } n \to \infty, \tag{6}$$

with probability one. Fix samples for which (3), (4) and (6) hold. We show $\hat{\theta}_n \to \theta^*$ for these samples. By (3),

$$F(\hat{\theta}_n) \leq \bar{S}_n(\hat{\theta}_n) \leq \bar{S}_n(\theta^*), \qquad \text{for } n \geq N.$$

Since $\bar{S}_n(\theta^*) \to E_*(S(X, \theta^*))$, the right side is bounded. Since $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space.

Suppose θ is a limit point of $\hat{\theta}_n$. Then $\hat{\theta}_n \to \theta$ along a subsequence. By (5),

$$\bar{S}_n(\theta) - \bar{S}_n(\theta^*) \leq \bar{S}_n(\theta) - \bar{S}_n(\hat{\theta}_n) \leq \delta(\theta, \hat{\theta}_n) \cdot \frac{1}{n}\sum_{k=1}^{n} g(X_k). \tag{7}$$

Since the left side of (6) converges to $E_*(g(X))$, and $\delta(\theta, \hat{\theta}_n) \to 0$ along a subsequence, the right side of (7) goes to zero along a subsequence. By (1) and (4), the left side of (7) converges to $I(\theta^*, \theta)$, hence $I(\theta^*, \theta) \leq 0$. But $I(\theta^*, \theta) \geq 0$, hence $I(\theta^*, \theta) = 0$, showing $\theta = \theta^*$. Thus $\hat{\theta}_n \to \theta^*$ as $n \to \infty$. ☐

4. Terminology History

The relative information $I(\theta^*, \theta)$ is often called "relative entropy" [5, 16]. The term "entropy," originating in thermodynamics, was coined by Clausius in his 1865 paper [4]. Boltzmann, in his 1872 paper [2], derived his H theorem, using a quantity H proportional to the negative of entropy. Gibbs, in his 1902 book [11], building on Boltzmann's 1877 combinatorial interpretation [3] and Planck's 1901 paper [18], reversed Boltzmann's sign convention, writing entropy with the sign convention used today.

Shannon, in his 1948 paper [21], independently defined

$$H(p) = -\sum_{i=1}^{d} p_i \log p_i.$$

Von Neumann had earlier defined quantum entropy in his 1932 book [17], and when Shannon consulted him around 1939–1940 about terminology for H, von Neumann suggested "entropy," which Shannon adopted [23]. In his remark "H is then, for example, the H in Boltzmann's H theorem," Shannon connects his H to Boltzmann's H, despite the difference in sign conventions.

The relative information $I(\theta^*, \theta)$ is jointly convex in the densities $p(X, \theta^*)$ and $p(X, \theta)$, as is apparent in the special case (2), whereas the entropy $H(p)$ above is strictly concave in p. From this perspective, the terminology "relative entropy" can seem counterintuitive. Indeed, based on curvature, one might argue that the *negative* of (2) is the more natural candidate for "relative entropy." In the Python `scipy.stats` package, the terminology also reflects the same distinction: `entropy(p)` returns the concave H, whereas `entropy(p,q)` returns the convex I.

This terminology can be understood by remembering the conceptual opposition between entropy and information: entropy is concave, whereas information is convex. The two notions are closely related, but differ by sign and by the role they play in optimization.

Similar remarks apply to the "cross-entropy" loss J of machine learning. Since J differs from I by a constant, one might analogously call J the *cross-information*, and its negative "cross-entropy." The above alternatives, however, are matters of terminology rather than mathematical substance.

While many authors use terminology different from ours, some align partially with our usage. Schervish, in his 1995 text [20], uses 'Kullback-Leibler information" for I, which is more descriptive than the widely used 'Kullback-Leibler divergence" [1, 15]. Kullback himself, in his 1959 monograph [14], refers to I as an "information function." Donsker and Varadhan, in their 1975 large deviations paper [8], denote their rate function by I rather than H; the relative information considered here is a special case of their I.

Doob's 1934 proof appeared at the dawn of abstract probability theory, within the milieu of Daniell, Fréchet, Hopf, Khintchine, Kolmogorov, Lévy, and Wiener, and within a year of Kolmogorov's 1933 foundational extension theorem [13].

5. Multivariate Normal

Let μ be in $\mathbf{R}^d$, Q a positive definite symmetric $d\times d$ matrix, and $\theta=(\mu,Q)$. Let $v\cdot w$ be the dot product in $\mathbf{R}^d$. With X in $\mathbf{R}^d$, the *normal density with mean* μ *and variance* Q is $p(X,\theta)=e^{-S(X,\theta)}$ where

$$S(X,\theta)=\frac{1}{2}\log\det(2\pi Q)+\frac{1}{2}(X-\mu)\cdot Q^{-1}(X-\mu). \tag{8}$$

This defines the parametric family.

A random variable X in $\mathbf{R}^d$ is *normally distributed with parameter* $\theta^*=(\mu^*,Q^*)$ if its moment-generating function is

$$E_*\left(e^{v\cdot X}\right)=\exp\left(v\cdot\mu^*+\frac{1}{2}v\cdot Q^*v\right). \tag{9}$$

Suppose X is normally distributed with parameter $\theta^*=(\mu^*,Q^*)$ as in (9), and let X_1, X_2, … be independent and identically distributed copies[4] of X. We show θ^* is the true parameter, and we verify **A** and **B**.

Assumption **B** follows from (8). To establish **A**, we need a few elementary facts about positive definite matrices. We state them as lemmas and prove them in the appendix. The proofs deliberately avoid invoking determinant multiplicativity or continuity properties of eigenvalues: the point is not merely to verify the required facts, but to expose the elementary mechanism behind them. These lemmas are useful in their own right and give a short, self-contained route to the properties of the multivariate normal surprisal.

Given vectors v, w in $\mathbf{R}^d$, the *tensor product* $v\otimes w$ is the matrix whose ij-th entry is v_iw_j. Then $v\otimes v$ is symmetric,

$$v\cdot Qv=\operatorname{trace}(Q(v\otimes v)),\qquad v\otimes w+w\otimes v\le\epsilon v\otimes v+\frac{1}{\epsilon}w\otimes w, \tag{10}$$

and, by the eigenvalue decomposition, every symmetric matrix is a linear combination of matrices of the form $v\otimes v$.

Given symmetric matrices Q, R, S, we say $Q\ge 0$ if $v\cdot Qv\ge 0$ for all v, and $Q\ge R$ if $Q-R\ge 0$. If $Q\ge\delta I$ for some $\delta>0$, we say $Q>0$. If $Q\ge 0$ and $R\ge S$, then $\operatorname{trace}(QR)\ge\operatorname{trace}(QS)$ and $\operatorname{trace}(QR)=\operatorname{trace}(RQ)$.

The determinant $\det(Q)$ is the product of the eigenvalues of Q.

Lemma 1. *With Q invertible symmetric and t a scalar,* $\det(Q+tv\otimes v)=\det(Q)\left(1+tv\cdot Q^{-1}v\right)$.

Lemma 2. *If $S>0$ and $Z>0$, then* $\det(SZS)=\det(Z)\det(S)^2$.

Lemma 3. *For c scalar, let $\epsilon=e^{-c}$ and $h(\lambda)=\log\lambda+1/\lambda$, $\lambda>0$. Then $h(\lambda)$ is minimized iff $\lambda=1$, and $h(\lambda)\le c$ implies $\epsilon\le\lambda\le 1/\epsilon$.*

[4] This setup is a special case of the Kolmogorov extension theorem.

Lemma 4. *For $Q^* > 0$ and c scalar, let $\epsilon = e^{-2c}\det(2\pi Q^*)$ and*

$$F_*(Q, Q^*) = \frac{1}{2}\log\det(2\pi Q) + \frac{1}{2}\mathrm{trace}(Q^{-1}Q^*), \qquad Q > 0. \tag{11}$$

Then $F_*(Q, Q^*)$ is minimized iff $Q = Q^*$, and $F_*(Q, Q^*) \le c$ implies $\epsilon Q^* \le Q \le Q^*/\epsilon$.

Let $f(\mu) = Q^* + (\mu^* - \mu) \otimes (\mu^* - \mu)$. Given $\theta = (\mu, Q)$, let

$$F(\mu, Q) = F_*(Q, f(\mu)) = F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu).$$

By Lemma 4, (μ^*, Q^*) is the unique minimizer of $F(\mu, Q)$.

The second conclusion of Lemma 4 is precisely the coercivity needed to establish properness. In (9), multiply both sides by $e^{-v\cdot\mu}$, then replace v by tv and differentiate twice at $t = 0$. This yields

$$E_*(X) = \mu^*, \qquad E_*\left((X - \mu) \otimes (X - \mu)\right) = f(\mu). \tag{12}$$

It follows $E_*(S(X, \theta)) = F(\mu, Q)$, and $\theta^* = (\mu^*, Q^*)$ is the true parameter.

If $F(\mu, Q) \le c$, then $F_*(Q, Q^*) \le c$, hence $\epsilon Q^* \le Q \le Q^*/\epsilon$, confining Q to a compact subset of the space of positive definite matrices. From this,

$$\begin{aligned} c &\ge F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu) \\ &\ge F_*(Q^*, Q^*) + \frac{\epsilon}{2}(\mu^* - \mu) \cdot (Q^*)^{-1}(\mu^* - \mu), \end{aligned}$$

confining μ to a compact subset of $\mathbf{R}^d$. Thus $F(\mu, Q)$ is proper.

With

$$\hat{\mu}_n = \frac{1}{n}\sum_{k=1}^{n} X_k, \qquad Q_n(\mu) = \frac{1}{n}\sum_{k=1}^{n}(X_k - \mu) \otimes (X_k - \mu), \qquad \hat{Q}_n = Q_n(\hat{\mu}_n),$$

let $\hat{\theta}_n = (\hat{\mu}_n, \hat{Q}_n)$.

Let $0 < \epsilon < 1/2$. By the LLN and (12) with $\mu = \mu^*$, there is $N \ge 1$, depending on the samples, such that, with probability one,

$$(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \le \epsilon^2 Q^*, \qquad \text{for } n \ge N,$$

and

$$Q_n(\mu^*) \ge (1 - \epsilon)Q^*, \qquad \text{for } n \ge N.$$

Let

$$f_\epsilon(\mu) = (1 - \epsilon)f(\mu) - \epsilon Q^*, \qquad F_\epsilon(\theta) = F_\epsilon(\mu, Q) = F_*(Q, f_\epsilon(\mu)).$$

Since $f_\epsilon(\mu) \ge (1 - 2\epsilon)f(\mu)$,

$$F_\epsilon(\mu, (1 - 2\epsilon)Q) \ge \frac{1}{2}\log\det((1 - 2\epsilon)I) + F(\mu, Q),$$

hence $F_\epsilon(\theta)$ is proper.

With $w = \hat{\mu}_n - \mu^*$ and $v = \mu - \mu^*$, using (10),

$$v \otimes w + w \otimes v \le \epsilon(\mu^* - \mu) \otimes (\mu^* - \mu) + \frac{1}{\epsilon}(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \le \epsilon f(\mu)$$

for all μ and $n \ge N$. Writing $X_k - \mu = (X_k - \mu^*) - v$,

$$Q_n(\mu) = Q_n(\mu^*) + v \otimes v - (v \otimes w + w \otimes v).$$

Combined with the above inequalities, this leads to

$$Q_n(\mu) \ge f_\epsilon(\mu), \qquad \text{for all } \mu \text{ and } n \ge N.$$

In particular, since $f_\epsilon(\mu) \ge (1 - 2\epsilon)Q^*$, we have shown $\hat{Q}_n > 0$ for $n \ge N$.

Since $\bar{S}_n(\theta) = F_*(Q, Q_n(\mu))$ and $Q_n(\mu) \ge \hat{Q}_n$ by least squares, by Lemma 4,

$$\bar{S}_n(\theta) = F_*(Q, Q_n(\mu)) \ge F_*(Q, \hat{Q}_n) \ge F_*(\hat{Q}_n, \hat{Q}_n) = \bar{S}_n(\hat{\theta}_n),$$

for all θ and $n \ge N$. Thus $\hat{\theta}_n$ is the MLE for $n \ge N$, and

$$\bar{S}_n(\theta) = F_*(Q, Q_n(\mu)) \ge F_\epsilon(\theta), \qquad \text{for all } \theta \text{ and } n \ge N,$$

establishing **A**.

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Appendix A. Proofs of Lemmas

The proofs below use only elementary linear algebra and calculus, and deliberately avoid determinant multiplicativity and continuity properties of eigenvalues. Besides supplying the facts needed in the multivariate normal example, the lemmas isolate several useful elementary properties of positive definite matrices.

Proof of (4) *with probability one.* By separability of the parameter space, there is a countable dense set D of parameters. Since D is countable, with probability one,

$$\bar{S}_n(\theta') \to E_*(S(X,\theta')), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta' \text{ in } D. \tag{13}$$

Fix samples for which (6) and (13) hold. We establish (4) for these samples. If θ, θ' are any parameters, by (5) and the triangle inequality,

$$\begin{aligned} |\bar{S}_n(\theta) - E_*(S(X,\theta))| \leq &|\bar{S}_n(\theta') - E_*(S(X,\theta'))| \\ &+ \delta(\theta,\theta') \left(E_*(g(X)) + \frac{1}{n}\sum_{k=1}^n g(X_k) \right). \end{aligned} \tag{14}$$

Now take the $\limsup_{n\to\infty}$ in (14). By (13),

$$\limsup_{n\to\infty} |\bar{S}_n(\theta) - E_*(S(X,\theta))| \leq 2\delta(\theta,\theta')E_*(g(X)), \quad \text{for all } \theta' \text{ in } D \text{ and all } \theta.$$

Now take the infimum of the right side over θ' in D. Since D is dense, $\delta(\theta,\theta')$ is continuous, and $\delta(\theta,\theta) = 0$, the right side is arbitrarily small, while the left side does not depend on θ'. This establishes (4). ☐

Proof of Lemma 1. Assume $t \neq 0$. If v is orthogonal to an eigenvector x of Q, then we reduce to a lower-dimensional case by restricting to $x^\perp$. Hence we may assume $v \cdot x \neq 0$ for all eigenvectors x of Q. If v is orthogonal to an eigenvector x' of $Q' = Q + tv \otimes v$, then x' is an eigenvector of Q, contradicting our assumption. Hence we may assume $v \cdot x' \neq 0$ for all eigenvectors x' of Q'. If x'_1 and x'_2 are linearly independent eigenvectors of Q' corresponding to the same eigenvalue, then, for suitable r_1, r_2, $x' = r_1 x'_1 + r_2 x'_2$ is an eigenvector of Q' satisfying $v \cdot x' = 0$, contradicting our assumptions. Hence we may assume the eigenvalues of Q' are simple. If x and x' are eigenvectors for Q and Q' corresponding to the same eigenvalue, then

$$t(v \cdot x')(v \cdot x) = x' \cdot (tv \otimes v)x = x' \cdot Q'x - x' \cdot Qx = 0,$$

contradicting our assumptions. Hence we may assume the eigenvalues of Q and the eigenvalues of Q' do not overlap. Let λ_1, λ_2, $\ldots$, λ_d be the eigenvalues of Q and λ'_1, λ'_2, $\ldots$, λ'_d the eigenvalues of Q'. Let v_1, v_2, $\ldots$, v_d be the corresponding orthonormal basis of eigenvectors of Q. Then the *secular equation* is

$$\sigma(z) = 1 + tv \cdot (Q - zI)^{-1}v = 1 + t\sum_{i=1}^d \frac{(v \cdot v_i)^2}{\lambda_i - z} = 0.$$

If λ' is an eigenvalue of Q' with eigenvector x', then $(Q' - \lambda' I)x' = 0$, hence $(Q - \lambda' I)x' + t(v \cdot x')v = 0$. Taking the dot product with $(Q - \lambda' I)^{-1}v$ yields $v \cdot x' + t(v \cdot x')(v \cdot (Q - \lambda' I)^{-1}v)$, hence $\sigma(\lambda') = 0$. If we let

$$p(z) = \det(Q - zI)\sigma(z),$$

then $p(z)$ is a polynomial and $p(\lambda') = 0$. Since λ'_1, λ'_2, $\ldots$, λ'_d are simple, $p(z) = \prod_{i=1}^d (\lambda'_i - z)$. Since the product of the roots is $p(0)$, the result follows. ☐

Proof of Lemma 2. Assume the result holds for Z. By Lemma 1,

$$\begin{aligned} \det(S(Z + v \otimes v)S) &= \det(SZS + Sv \otimes Sv) \\ &= \det(SZS)(1 + Sv \cdot (SZS)^{-1}Sv) \\ &= \det(Z)\det(S)^2(1 + v \cdot Z^{-1}v) = \det(Z + v \otimes v)\det(S)^2, \end{aligned}$$

hence the result holds for $Z + v \otimes v$. If $Z = \delta I$, the result holds. Since every $Z > 0$ is a sum of δI and a sum of matrices of the form $v \otimes v$, the result follows by induction over the number of terms in the sum. ☐

Proof of Lemma 3. The first part is checked by differentiation. For the second part, clearly $h(\lambda) \leq c$ implies $\lambda \leq 1/\epsilon$. Since $\lambda \log \lambda \geq -1/e$ and $(e-1)e^c \geq ec$ for $c \geq 1$, $h(\lambda) \leq c$ and $c \geq 1$ implies $\lambda \geq (1-1/e)/c \geq \epsilon$. ☐

Proof of Lemma 4. Since $Q^* > 0$, there is $S > 0$ with $Q^* = S^2$. Let $Z = S^{-1}QS^{-1}$ and let $\lambda_1, \lambda_2, \ldots, \lambda_d$ be the eigenvalues of Z. Then $\operatorname{trace}(Q^{-1}Q^*) = \operatorname{trace}(SQ^{-1}S) = \operatorname{trace}(Z^{-1})$. By Lemmas 2 and 3,

$$\begin{aligned} 2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) &= \log\det Q - \log\det Q^* + \operatorname{trace}(Q^{-1}Q^* - I) \\ &= \log\det(SZS) - \log\det(S)^2 + \operatorname{trace}(Z^{-1} - I) \\ &= \log\det Z + \operatorname{trace}(Z^{-1} - I) \\ &= \sum_{i=1}^d (h(\lambda_i) - 1) \geq 0, \end{aligned} \tag{15}$$

establishing $F_*(Q, Q^*) \geq F_*(Q^*, Q^*)$. If (15) is an equality, $Z = I$ hence $Q = Q^*$. Now suppose $F_*(Q, Q^*) \leq c$. By (15), $h(\lambda_i) \leq \log(1/\epsilon)$ for $i = 1, 2, \ldots, d$. By Lemma 3, $\epsilon \leq \lambda_i \leq 1/\epsilon$ for $i = 1, 2, \ldots, d$. This implies $\epsilon I \leq Z \leq I/\epsilon$, hence $\epsilon Q^* \leq Q \leq Q^*/\epsilon$. ☐

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